THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pH of a solution is 5. If its pH becomes 3, its H^+ concentration: 1

(a) increases 2 times **(b)** increases 100 times
(c) decreases 100 times **(d)** decreases 2 times

2. The solid residue obtained on heating washing soda crystals is: 1

(a) anhydrous sodium carbonate **(b)** sodium chloride
(c) sodium hydroxide **(d)** sodium bicarbonate

3. A student tests an unknown solution with litmus and gets no change. The solution is most likely: 1

(a) strongly acidic **(b)** basic **(c)** acidic **(d)** neutral

4. Which of the following will have the highest H^+ ion concentration? 1

(a) pH 1 **(b)** pH 10 **(c)** pH 4 **(d)** pH 7

5. $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ is a: 1

(a) displacement **(b)** combination
(c) neutralisation **(d)** thermal decomposition

6. Which gas is evolved when $NaOH$ reacts with zinc? 1

(a) chlorine **(b)** oxygen **(c)** hydrogen **(d)** carbon dioxide

7. Which of the following pairs of solutions will produce a neutral salt solution? 1

(a) HCl and NaOH **(b)** H_2CO_3 and NaOH **(c)** HCl and NH_4OH **(d)** CH_3COOH and NaOH

8. On passing excess CO_2 in lime water, the milkiness: 1

(a) disappears **(b)** turns pink **(c)** increases **(d)** remains same

9. Which of the following is a strong base? 1

(a) $Al(OH)_3$ **(b)** $Cu(OH)_2$ **(c)** NaOH **(d)** $Mg(OH)_2$

10. The salt CH_3COONa in water is: 1

(a) amphoteric **(b)** basic **(c)** acidic **(d)** neutral

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between a strong acid and a weak acid with one example each. 2

12. How does a farmer treat acidic soil? Give reason. 2

13. Why is baking powder used in making cakes soft and spongy? 2

14. A solution has pH 2 and another has pH 4. Compare their H^+ concentrations. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student adds Mg ribbon to three test tubes with HCl, CH_3COOH and distilled water. Predict the observation in each and explain. 3

16. Explain the preparation of sodium hydroxide, naming the electrodes and products. 3

17. Why does vinegar remove limescale (calcium carbonate)? Write the reaction. 3

OR

Explain how a solution of pH 7 can be made from pH 1 and pH 13 solutions, giving the principle. 3

18. State what happens when plaster of Paris is mixed with water and explain why a cast hardens. 3

OR

How can you distinguish between a solution of NaOH and Na_2CO_3 using HCl? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student has four unlabelled test tubes with HCl, NaOH, NaCl and Na_2CO_3 solutions. Suggest tests to identify each. 5

OR

Explain the preparation and setting of plaster of Paris with equations. State its uses. 5

20. Describe the properties and formulae of acids and bases using equations: reaction with metals, carbonates and each other. 5

OR

Describe an experiment to show the presence of water of crystallisation in copper sulphate crystals. 5

END OF QUESTION PAPER — Total: 40 Marks

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